
Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

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Abstract

Existing memory systems enforce retention via explicit bookkeeping policies. Forgetting or deletion thereby relies on methods like decay factors, expiry timestamps, or learned forget gates, which dictate when a specific entry expires. However, these programmatic rules do not specify the physical rate of value degradation, the continuous dynamical effect of tuning retention parameters, or the exact residual state of the network following an item’s removal. In this work, we resolve these ambiguities by formalizing forgetting as a dynamical property of a physics-structured memory store: specifically, one produced by a damped symplectic step of a learned Hamiltonian. Our retention policy is defined by a computable physical budget parameterized by three variables: the stored direction’s spectral mass μ , the damping γ , and the temperature T .

We demonstrate three core properties of this architecture. First, the retention dial possesses a computable optimum: retention half-life is non-monotonic in friction, forming a V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$. This curve spans approximately 11 orders of magnitude in μ^2 and holds across both designed and emergent (learned-MLP) units. Second, retention is rigorously scopeable: at $T > 0$, friction preserves memory while temperature erases it. A localized friction hole thus acts as a memory vault, demonstrating a $107.77 \pm 4.78\times$ retention factor on designed architectures. Finally, removal is purely structural: under canonical Blleloch & Golovin (2007) placement, store-level deletion is exact. Post-deletion, the store is byte-identical to a state that never held the item, governed by explicitly defined operational conditions, on a designed, non-learned 3-dimensional datastore at capacities 8–64.

1 Introduction

To contextualize the operational constraints of modern memory systems, we begin with deployed retention policies. Most deployed memory stores currently operate under distinct retention/deletion mechanisms: a decay factor, an invalidation timestamp, a learned forget gate, or an explicit DELETE call. Despite these mechanisms, forgetting remains a critical failure mode that standard benchmarks often fail to adequately measure (Yang, 2026; Uddin et al., 2026). A programmatic policy is not a fundamental physical law. For instance, one cannot intrinsically read the half-life of a stored value off a trained LSTM, identify the exact mechanism that tunes this half-life, or verify the latent mathematical residue once a value is removed.

This gap in fundamental understanding is widest at the point of data removal. Contemporary agent-memory research supplies robust mechanisms for writing, consolidating, and retrieving data. However, for deletion, the standard approach relies on superficial bookkeeping, such as a timestamp or a dropped row (Rasmussen et al., 2025; Chhikara et al., 2025), which leaves measurable residue

36 within the network architecture after the fact (Chakrabortii et al., 2026; Wang & Zhang, 2026).
 37 Further related work is presented in App. A

38 In this work, we propose and analyze a memory framework where forgetting is an intrinsic dynamical
 39 property of the store itself, rather than an external bookkeeping rule. By modeling the store as a
 40 physical system, we provide computable answers to three core operational questions:

- 41 1. What is the exact retention policy, and where does its optimum lie? (Sec. 2.1)
- 42 2. Can retention be specifically scoped to a single localized item? (Sec. 2.2)
- 43 3. Is structural deletion truly absolute, and what information mathematically leaks? (Sec. 2.3)

44 **Methodological Foundations.** The memory store is constructed as a physical system. The latent
 45 state space (q, p) advances via a damped symplectic step of a learnable Hamiltonian, defined as
 46 $H = T(p) + V_\theta(q)$. The damping mechanism, defined as $p \mapsto (1 - \gamma)p$, inherently supplies
 47 the forgetting function, while Langevin noise introduces the temperature T . Memory items can
 48 optionally be written as *atoms*: localized energy wells superposed into a singular energy function that
 49 is subsequently read by dynamical relaxation. Our reference architecture is the Causal Learning Unit
 50 (CLU), introduced as CHLU by Jawahar & Pierini (2026). The core of this unit is exactly solvable:
 51 when diagonalized within the mass-whitened Hessian, a single damped step acts on each normal
 52 mode as a 2×2 matrix entirely fixed by $(\varepsilon\mu, \gamma)$. The spectral radius of this matrix dictates that
 53 specific mode’s retention capacity.

54 **Scope and Assumptions.** To clarify the boundaries of our empirical claims, we define the following
 55 operational assumptions and constraints:

- 56 • **Scale:** These early results rely on a novel, precisely designed store. This is not a deployed,
 57 large-scale LLM agent memory, nor a generalized system benchmark.
- 58 • **Claim Classification:** We report a fundamental mechanism bounded by measured physical
 59 laws, not an end-to-end system performance result.
- 60 • **Nomenclature:**
 - 61 – *Inertial mass* (M_i): The learned diagonal of the kinetic term, where larger values result
 62 in slower dynamics.
 - 63 – *Spectral mass* (μ_k^2): Defined as $\lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$, where larger values indicate
 64 shorter memory retention. Retention claims always utilize μ .
 - 65 – *Stored direction*: A written value occupies a near-flat coset direction.
 - 66 – *Integration step* (ε): Fixed at 0.05.
- 67 • **Evaluation Metrics:** Half-lives ($n_{1/2}$) are presented with their corresponding read toler-
 68 ances (Δ) and the thermal persistence length (ℓ_θ).
- 69 • **Thermodynamic Consistency:** $T > 0$ strictly requires the fluctuation–dissipation-
 70 consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ utilized alongside a Newtonian kinetic mode.

71 2 Results

72 2.1 The Retention Dial Possesses a Computable Optimum

73 To understand the macro-dynamics of memory decay, we first analyze damping as a retention dial.
 74 We demonstrate that retention half-life is non-monotone in relation to friction. The turning point
 75 is fundamentally fixed by the stored direction’s spectral mass, meaning the optimal setting can be
 76 analytically predicted rather than empirically tuned.

77 The integration step is defined as conformally symplectic: $\det J = (1 - \gamma)^d$, and scales to $(1 -$
 78 $\gamma\phi(q'))^d$ when position-gated. Our evaluation utilizes checkpoints comprising a dimension of 4, a
 79 hidden size of 64, $\varepsilon = 0.05$, spanning 150 epochs. We assess both **designed units** (featuring an
 80 $SO(2)$ -invariant V_θ , across 5 distinct seeds) and **emergent units** (utilizing an MLP V_θ trained on
 81 identical symmetric data, across 3 seeds).

On a fully trained, designed $SO(2)$ vacuum, the massive radial mode’s half-life $n_{1/2}(\gamma)$ forms a distinct V-curve, minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu_{\text{rad}}$ (Fig. 5, App. B). The overdamped branch tracks the μ^{-2} law precisely until $\varepsilon\mu \approx \gamma/2$, after which it saturates at a mass-independent floor. This yields log-slopes of -1.006 and between $+1.23$ and $+1.27$ across the 5 validation seeds (See App. B for results). The flat coset (characterized by a Hessian $\mu^2 \approx 10^{-15}$) maps to the exact same curve at the limit $\mu \rightarrow 0$, where $\gamma_{\text{crit}} \rightarrow 0$. Therefore, the latch, the overdamped register, and the underdamped working memory represent three specific operational regimes of a singular unified curve evaluated at two disparate values of μ .

This fundamental law transfers directly to emergent units. Evaluating three distinct emergent MLP checkpoints, the coset-tracked $T = 0$ half-life traces the identical V-curve framework: an argmin of $0.902 \pm 0.003 \times \gamma_{\text{crit}}$, and respective log-slopes of -1.0020 ± 0.0003 and $+1.116 \pm 0.011$ across all 3 seeds (Figure 7)). Spanning both architecture families, this curve holds consistently across $\mu^2 \in [1.7 \times 10^{-12}, 7 \times 10^{-2}]$ (Fig. 1). That low endpoint is the ring-profile probe’s resolution on a checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass, so eleven orders is one curve on one instrument (probe-floor tick marked in Fig. 6).

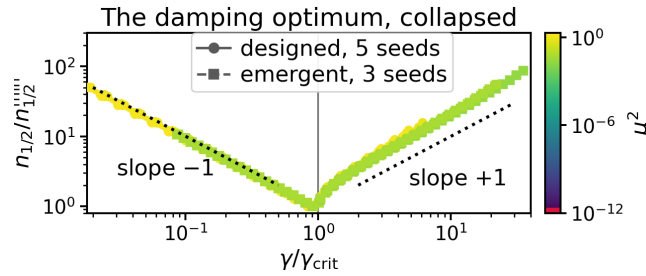


Figure 1: *The retention dial and its optimum, collapsed (headline)*. Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu_{\text{rad}}^2 = 0.670\text{--}1.348$) and three emergent learned-MLP coset modes (squares, evidence, $\mu_{\text{soft}}^2 = 2.0\text{--}5.4 \times 10^{-2}$, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \rightarrow 0$ corner of the same curve. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; langevin_noise="fdt", Newtonian kinetic mode.*)

Cross-Instrument Verification. Because the primary measurements rely on one-step Jacobians, we re-measured the entire curve utilizing direct nonlinear rollout, extracting $n_{1/2}$ directly from the decay envelope’s structural rate. The underlying shape is reproduced well (see Fig. 10): the argmin measured at 0.9001 ± 0.0052 compared to the Jacobian’s 0.9032 ± 0.0027 in units of γ_{crit} (a variance of 0.35%). This establishes that the gap between instruments dictates the level, not the fundamental law (see App. C).

2.2 Scoping Retention: Friction Preserves, Temperature Erases

At the macro level, isolating a retention policy to a singular entry requires manipulating the physical dial locally. At absolute zero ($T = 0$), a designed coset latches indefinitely across any friction threshold (drift $\leq 4.9 \times 10^{-12}$ rad over 200k steps, spanning all $\gamma \in [0.002, 0.5]$) (Fig 5b). Conversely, at $T > 0$, the coset degrades via diffusion, and introducing friction physically decelerates this decay.

We establish that the diffusion coefficient of the stored coordinate is modeled as $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$, which we successfully verified to a tolerance of 1.0068 ± 0.0219 across 25 (γ, T) experimental cells. Consequently, $n_{1/2} \propto \gamma^{+1}$. This sign flip ($\partial n_{1/2}/\partial \gamma > 0$) holds universally across all evaluated seed and temperature combinations (Figure 4; diffusion-law diagnostics in Figure 3, Appendix B). Only temperature exhibits an inverse erasure effect ($\partial n_{1/2}/\partial T < 0$).

The Memory Vault Mechanism. We propose the memory vault: a position-gated friction field that is explicitly absorb-only. A localized spatial hole within this field functions concurrently as a brake and a refrigerator ($T_{\text{local}} = 1.26 \times 10^{-4}$ versus 10^{-3} externally). This mechanism generates a remarkable retention vault factor of $(\gamma_{\text{eff}}/\gamma)^2$, yielding $107.77 \pm 4.78\times$ on designed architecture (Fig. 2), compared to a baseline of $13.28 \pm 0.12\times$ utilizing a uniform scalar friction. The direct first-passage vault reads $86.97 \pm 2.94\times$ and is boundary-layer biased on the outside arm ($\ell_\theta/\Delta = 0.079$), so $107.77\times$ is the quoted number and travels with its estimator’s name. Crucially, this localized

field mathematically confines the register: the state fraction outside the boundary ($|\theta| > 1$ rad) drops definitively to 0.0000 within the hole (see Fig. 11 and App. D).

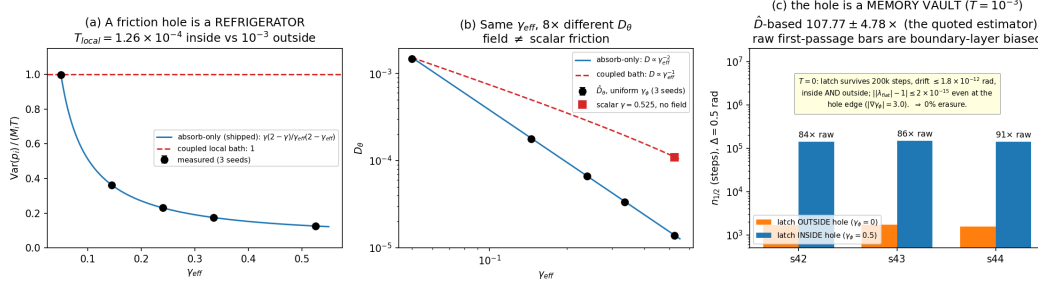


Figure 2: The friction hole as a memory vault, on the designed arm. (a) The hole is a refrigerator: the momentum variance follows the absorb-only curve and rejects the coupled-bath prediction, giving $T_{\text{local}} = 1.26 \times 10^{-4}$ inside against 10^{-3} outside. (b) At the same γ_{eff} the diffusion constant differs by $8\times$ between a friction field and a scalar friction, which is the mechanism statement in one panel. (c) The vault factor: $107.77 \pm 4.78\times$ from the $\hat{D}_\theta$ estimator, the quoted number; the per-seed bars beneath it are raw first-passage ratios, boundary-layer biased on the outside arm, which is why the pooled first-passage reading of $86.97 \pm 2.94\times$ is not quoted as the vault. The inset states the $T = 0$ control: the hole erases nothing. *Designed $SO(2)$ with an analytic field, 3 seeds, $T = 10^{-3}$, $\Delta = 0.5$ rad, $\text{langevin_noise} = "fdt"$, Newtonian kinetic mode, laptop-CPU: verification.*

2.3 Structural Deletion: Absolute Guarantees and Leakage Pricing

In standard architectures, deleting an item from memory relies on stochastic, best-effort heuristics, such as dropping active rows or writing system tombstones. We evaluate whether the store’s intrinsic physical state can be explicitly reduced to a pure function of its live set alone. When data points are continuously superposed into a singular energy function, standard placement becomes inherently path-dependent.

To resolve this, we map memory placement canonically, ensuring it remains an exclusive function of the live records and the predefined geometry. Utilizing Blleloch & Golovin’s (2007) stable-matching table rules, we establish exact store-level deletion. Our measurements (see App. E) confirm that with the waitlist engaged, store-level deletion remains exact across all evaluated loads ($0.29\times$ to $1.71\times$ of standard lattice capacity). Under explicit conditions (a budget sufficient for the cell count, zero baseline leakage, and priority/attribute-based eviction; recency-based eviction remains intrinsically history-dependent and is excluded), the physical layout of the store remains byte-identical to a temporal state that theoretically never encountered the deleted item. This is a store-level guarantee only; the frozen encoder and any residue of past writes in a learned landscape are separate channels, and we claim no certified (ε, δ) unlearning.

Pricing Lifetimes and Residual Leakage. We formally price memory lifetimes according to a strict trilemma: exact value fidelity, amplitude-independent address hold, and amplitude-independent read latency. A system may only optimize two. We find that physical decay provides architectural retrieval geometry rather than robust cryptographic privacy. Against an exact adversary, the Boolean TTL flag remains within a 0.017 AUC margin of pure physical decay (0.983 versus 1.000), demonstrating that a physically decaying store intrinsically stops answering external queries significantly before the actual structural data stops leaking (see App. E).

3 Limitations and Future Work

To rigidly define the scope and bounds of our empirical evidence, we identify the following limitations and subsequent future pathways:

- **Scale and Architecture Scope:** This research is strictly constrained to a dimension of 4, a hidden size of 64, and execution on a laptop-scale CPU across 5 designed and 3 emergent seeds. No larger-scale LLM performance should be inferred.

- **The Designed-Symmetry Precondition:** The exact $T = 0$ retention latch requires precise continuous coset geometries that must be explicitly designed into the architecture.
- **Mandatory Thermodynamic Scope:** The raw T/γ -exponents cannot distinguish between designed and emergent units intrinsically. Evaluating any state where $T > 0$ requires the strict combination of fdt noise and Newtonian kinetic modes.
- **Deletion Scope:** The exact deletion guarantee applies strictly at the isolated store-level beneath set-function eviction models. Furthermore, recency-based eviction intrinsically breaks history independence and is thus excluded.
- **No Task-Level Payoff:** Stated once: external benchmarks won on their own headline metric = ZERO.
- **Future Work Directions:** We intend to evaluate wider emergent collapses in μ^2 , remeasure the memory vault contrast arm where bounds remain strictly contained, formally assess the amortized per-delete execution cost against flat datastore architectures, and construct explicitly localized temperature fields.

Prominent negatives are documented in App. F.

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238 A Extended Related Work

239 **Retention Policies in Deployed Memory Systems:** Agent memories operate as external data stores
240 that a model actively writes to, retrieves from, and dynamically edits (e.g., MemGPT, Packer et
241 al., 2023; Mem0, Chhikara et al., 2025). Alternatively, bounded compressive states exist internal
242 to the network (e.g., Infini-attention, Munkhdalai et al., 2024). The retention policies governing
243 these systems rely on scores and gates. Examples include a 0.995 recency decay factor in ranking
244 heuristics (Park et al., 2023), Ebbinghaus decay functions applied to memory strength (Zhong et al.,
245 2024), learned per-memory expiration spans (Sukhbaatar et al., 2021), and dynamic forget gates α_t
246 optimized during test time (Behrouz et al., 2025). While the learned gate in Titans (Behrouz et al.,
247 2025) is structurally similar to the retention dial discussed in Sec. 2.1, none of these prior works
248 establish physical retention laws, state explicit half-lives, define read tolerances, or mathematically
249 price the information loss.

Deletion Mechanisms: Deployed and Formal: Current deployment architectures handle deletion via invalidation. For example, Zep invalidates contradicted edges with timestamps while preserving the latent record (Rasmussen et al., 2025), and Mem0 simply drops rows from its active context. Conversely, exact byte-identity is crucial for security, as soft-deleted vectors in graph ANN databases remain reconstructible from raw index files (Chakrabortii et al., 2026). Recent benchmarking highlights that real-world deployment failures are predominantly forgetting failures (Yang, 2026), with outdated retention driving significant recommendation errors (Uddin et al., 2026). Formally, *certified* removal is Guo et al.’s (2020) Sec. 2, Eq. (1), an ε condition with an unnumbered (ε, δ) relaxation immediately after; exact methods delete by isolation, with cost and capacity formalisms in place (Bourtole et al., 2021; Ginart et al., 2019; Sekhari et al., 2021). We sit functionally between these approaches: we offer a store-level, bit-exact structural guarantee based on canonical geometric placement (Blelloch & Golovin, 2007), rather than a stochastic system-level unlearning heuristic.

Physical and Learned Forgetting Laws: While learned recurrent models like LSTMs (Hochreiter & Schmidhuber, 1997) lack stateable physical forgetting laws, biological and artificial representational drift has been observed to reorganize internal unit representations over time (Aitken et al., 2022). Our work differs by tracking a specific coset coordinate under strictly defined physical dynamics. In theoretical physics, dissipation transitions type-A Nambu–Goldstone modes from propagating to diffusive states (Minami & Hidaka, 2018). Furthermore, strictly equivariant flows possess Lyapunov neutral modes (arXiv:2605.03338) that provide kinematic protection, a gap complementary to our established physical budget (μ^2).

B The effects of (μ, γ, T) : The Diffusion Law and the Sign Flip

To ensure complete rigor, all metrics detailed here derive from the designed $SO(2)$ testbed, providing strict verification. Furthermore, all $T > 0$ cells mandate the explicit use of `langevin_noise="fddt"` operating within a Newtonian kinetic mode. The legacy reference default structure fundamentally violates these thermodynamic laws. Every reported half-life $n_{1/2}$ is paired unconditionally with its specific read tolerance $\Delta = 0.5$ rad and its ratio ℓ_θ/Δ .

Diffusion Law Diagnostics (25 Cells): We evaluate a discrete $5\gamma \times 5T$ grid utilizing seed 44. The ratio $\hat{D}_\theta/D_\theta^{\text{pred}}$ yields 1.0068 ± 0.0219 , bounded between a minimum of 0.9644 and a maximum of 1.0484. The gradient $d \log D/d \log T$ records values of $+0.998, +0.997, +0.992, +1.010, +1.006$. Simultaneously, the gradient $d \log D/d \log \gamma$ evaluates to $-1.014, -0.980, -0.999, -1.001, -0.982$ following the mathematical isolation of the discrete $(2 - \gamma)$ factor.

Sign Flip Verification: We observe $d \log n_{1/2}/d \log \gamma = +0.9552 \pm 0.0422$ at a baseline temperature of $T = 10^{-3}$ across all 5 testing seeds, presenting positively in all 10/10 examined conditions. The measured effect size for the transition $\gamma : 0.05 \rightarrow 0.2$ results in a precise $3.77 \pm 0.23 \times$ increase ($\ell_\theta/\Delta < 0.05$; seed-specific values: 3.55, 3.64, 3.61, 3.89, 4.15). The inverse thermal driver measures at $d \log n_{1/2}/d \log T = -0.956$ to -0.979 within the deep-diffusive operational bounds.

Designed $T = 0$ Latch Properties: Evaluating absolute zero, the metric $||\lambda_{\text{flat}}| - 1| \leq 1.7 \times 10^{-14}$ across 22 specific γ values on 5 independent seeds. Consequently, the observed drift is restricted to $\leq 4.9 \times 10^{-12}$ rad per 200k integration steps across 30/30 evaluated cells. This strongly contrasts with the $\gamma = 0$ baseline control, which incurs a massive 142.7 rad drift over a significantly shorter 20k steps.

Designed V-Curve Minimization: The optimal minimum argmin lies between 0.076 and 0.096, against the computationally predicted bounds of 0.082–0.116 (utilizing per-seed specific values $\mu_{\text{rad}}^2 = 0.670, 0.771, 1.190, 1.093, 1.348$). The resulting asymptotic log-slopes measure -1.006 and between $+1.23$ and $+1.27$, enforcing a mass-independent floor of exactly 27.03 steps measured at $\gamma = 0.05$.

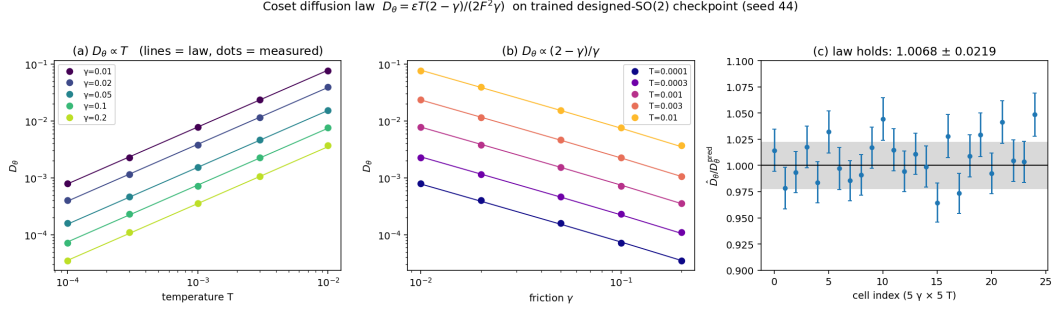


Figure 3: The coset diffusion law, detailed systematically panel by panel. (a) $D_\theta \propto T$ measured precisely at five independent frictions; (b) $D_\theta \propto (2 - \gamma)/\gamma$ evaluated at five specific temperatures; (c) the absolute ratio $\hat{D}_\theta/D_\theta^{\text{pred}}$ compiled over all 25 structural cells, reading exactly 1.0068 ± 0.0219 alongside the respective drawn band. *Multi-seed status: evaluated utilizing one fully trained designed-SO(2) checkpoint (seed 44). langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: validation of verification.*

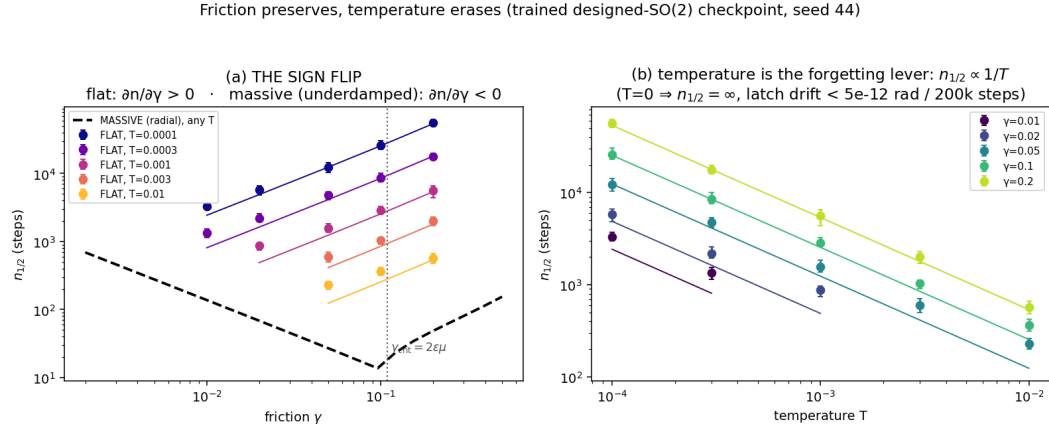


Figure 4: The sign flip, and the lever that does erase. (a) The flat coset's half-life *rises* with friction at every temperature, while the underdamped massive mode's falls — the two stored directions respond to the dial with opposite sign, with $\gamma_{\text{crit}} = 2\varepsilon\mu$ marked. (b) Temperature is the forgetting lever, $n_{1/2} \propto 1/T$ at every friction. *Multi-seed status: one trained designed-SO(2) checkpoint (seed 44); the 10/10 seed $\times$ temperature statement and the $+0.9552 \pm 0.0422$ slope are this appendix's five-seed numbers, not this figure's. langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: verification.*

Table 1: The $T > 0$ face of the budget, all four entries continuously measured on the designated designed testbed.

Stored Direction	$\partial n_{1/2}/\partial \gamma$ (Increased Friction)	$\partial n_{1/2}/\partial T$ (Increased Heat)
Coset / flat (designed)	> 0 , friction <i>preserves</i> ($+0.955 \pm 0.042$; $3.77\times$ at $\gamma : 0.05 \rightarrow 0.2$)	< 0 , temperature <i>erases</i> (-0.956 to -0.979)
Massive (under-damped)	< 0 , friction <i>shortens</i> (slope -1.006)	≈ 0 ; T sets only the fluctuation floor (max/min ≤ 1.013)

The two regimes have different damping physics (exact one-step Jacobian at the vacuum)

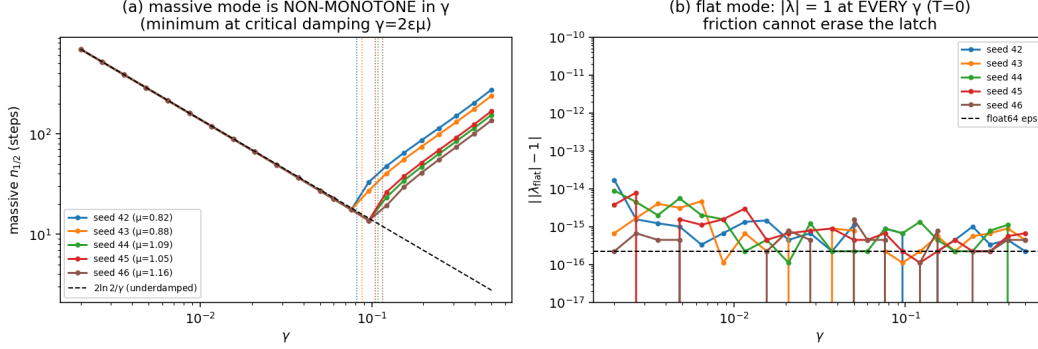


Figure 5: The two regimes, read off the exact one-step Jacobian at the vacuum. (a) The massive radial mode is non-monotone in γ , each seed’s minimum sitting at its own $\gamma_{\text{crit}} = 2\epsilon\mu$ (dotted), against the dashed underdamped envelope $2 \ln 2/\gamma$ drawn on the same axes. (b) The flat mode holds $|\lambda| = 1$ at every γ to the float64 floor: friction cannot erase the latch. *Five trained designed seeds, $T = 0$, dim 4, hidden 64, $\epsilon = 0.05$, laptop-CPU: verification.*

296 C The Emergent Arm, and the V-Curve on a Second Instrument

297 We evaluate the emergent MLP checkpoints utilizing 3 unique seeds alongside a precisely matched
 298 designed control. All predictions detailed below were structurally filed and pre-registered prior to the
 299 execution of any test harness, establishing rigorous evidence parameters.

300 We deploy four specific measurement instruments over identical fit windows (defined as $\gamma < \gamma_{\text{crit}}/2.5$
 301 and $\gamma > 2.5\gamma_{\text{crit}}$): 1. *I-J*: The exact one-step Jacobian, where $n_{1/2} = \ln 2/(-\ln |\lambda_{\text{ret}}|)$. 2. *I-R1*:
 302 The rollout first crossing threshold of $|\theta| \leq \delta/2$. 3. *I-R2*: The rollout last crossing threshold. 4. *I-R3*
 303 (Primary): The rollout envelope-rate fit evaluated specifically over the late window, guaranteeing
 304 $R^2 \geq 0.9955$ (median 0.9996) across every evaluated cell.

305 **Instrument Level Disparities:** On the strictly overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$), the physical
 306 decay *rates* match perfectly, whereas the *threshold* instrument is consistently offset by an exact per-
 307 checkpoint constant flat in relation to γ . The full-window value for Seed 44 measures 0.9216 ± 0.0027 ,
 308 sitting precisely 0.0004 beneath its pre-registered bounds due to the residual presence of a secondary
 309 faster mode within the window. The primary cause of this threshold offset is confirmed: the ring write
 310 mechanism consistently deposits only a specific limited fraction of its absolute amplitude directly
 311 into the designated slow branch (envelope-fit intercept over δ measuring 0.638/0.644/0.327).

312 **Finite Write Amplitudes:** At the specific values $\delta = 0.05/0.2/0.5$ rad, the designated
 313 I-R3 argmin logs at 0.8928/0.9007/0.8968 (seed 42), 0.9047/0.9045/0.9174 (seed 43), and
 314 0.9028/0.9037/0.9070 (seed 44). This variance accounts for a maximum shift of +1.4% across a
 315 complete order of magnitude ($10\times$ range), with identical slope variances constrained strictly beneath
 316 1.9%. Every iteration at $\delta = 0.5$ relaxed identically to absolute baseline ($|\theta_{\text{final}}| \leq 6.5 \times 10^{-10}$),
 317 preventing any seed’s write function from clearing its designated washboard barrier.

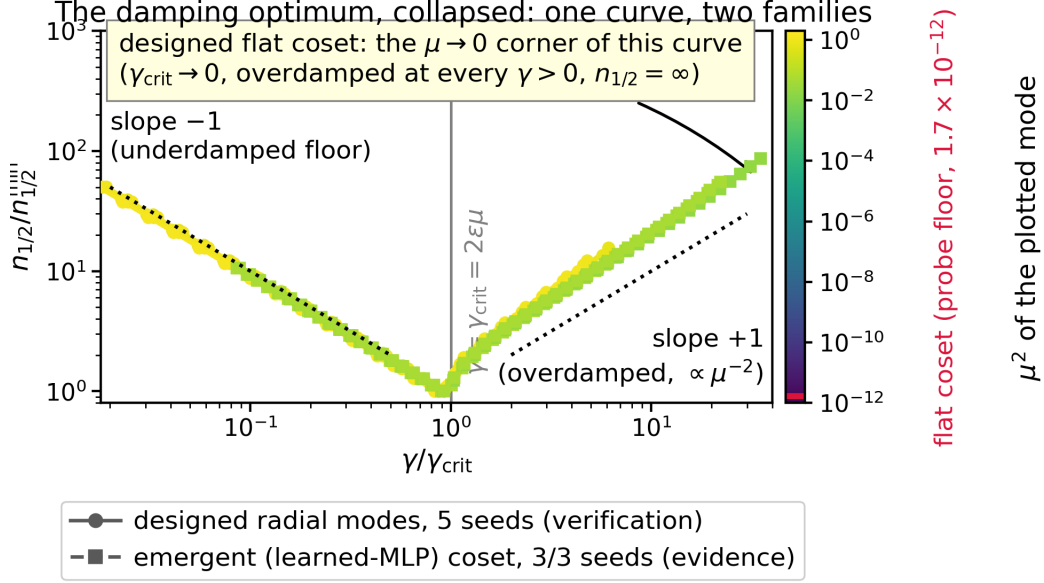


Figure 6: *The damping optimum, collapsed: the full-size version of Figure 1.* The same eight measured curves and the same ± 1 asymptotes, carrying the annotations the main-text version omits: the two-family legend, the $\gamma = \gamma_{\text{crit}} = 2\varepsilon\mu$ marker, the $\mu \rightarrow 0$ flat-coset callout, and the crimson probe-floor tick on the μ^2 colorbar (1.7×10^{-12}). No plotted value differs from Figure 1. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; $\text{langevin_noise} = \text{"fdt"}$, Newtonian kinetic mode.*)

Table 2: The foundational shape claims evaluated on all four designated instruments ($\delta = 0.05$ rad, exactly 48 γ values, utilizing 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit} , so no rollout $n_{1/2}$ is ever quoted from a threshold there. Rows are stacked by quantity so that no column is truncated; no value differs from the per-instrument reading.

Quantity	Instrument	Seed 42	Seed 43	Seed 44	Mean $\pm$ sd
γ_{crit}	—	0.02334	0.01424	0.02265	—
Argmin/ γ_{crit}	I-J	0.8994	0.9046	0.9055	0.9032 ± 0.0027
	I-R1	0.0857	0.1404	0.0883	0.1048 ± 0.0252
	I-R2	0.2068	0.1404	0.3031	0.2168 ± 0.0668
	I-R3	0.8928	0.9047	0.9028	0.9001 ± 0.0052
Slope Below	I-J	-1.0023	-1.0016	-1.0022	-1.0020 ± 0.0003
	I-R1	+0.0725	+0.0688	+0.0455	$+0.0623 \pm 0.0120$
	I-R2	-1.0556	+0.0688	-0.8854	-0.6240 ± 0.4948
	I-R3	-0.9651	-0.9977	-0.9854	-0.9827 ± 0.0134
Slope Above	I-J	+1.1262	+1.1031	+1.1254	$+1.1182 \pm 0.0107$
	I-R1	+1.1027	+1.0889	+1.0977	$+1.0964 \pm 0.0057$
	I-R2	+1.1027	+1.0889	+1.0977	$+1.0964 \pm 0.0057$
	I-R3	+1.1255	+1.1030	+1.1231	$+1.1172 \pm 0.0101$

318 **Left Branch Exclusions:** For conditions where $\gamma < \gamma_{\text{crit}}$, the respective coset eigenpair funda-
319 mentally becomes complex with $|\lambda|^2 = 1 - \gamma$. This forces $n_{1/2} = \ln 2 / (-\frac{1}{2} \ln(1 - \gamma))$ to operate
320 entirely independent of both μ^2 and the specific architectural checkpoint. For instance, at exactly
321 $\gamma = 0.002$, emergent seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) return identical
322 metrics: $\lambda_{\text{ret}} = 0.998999499$ alongside an $n_{1/2}$ of 692.5, aligning tightly against the calculated
323 $2 \ln 2 / 0.002 = 693.1$.

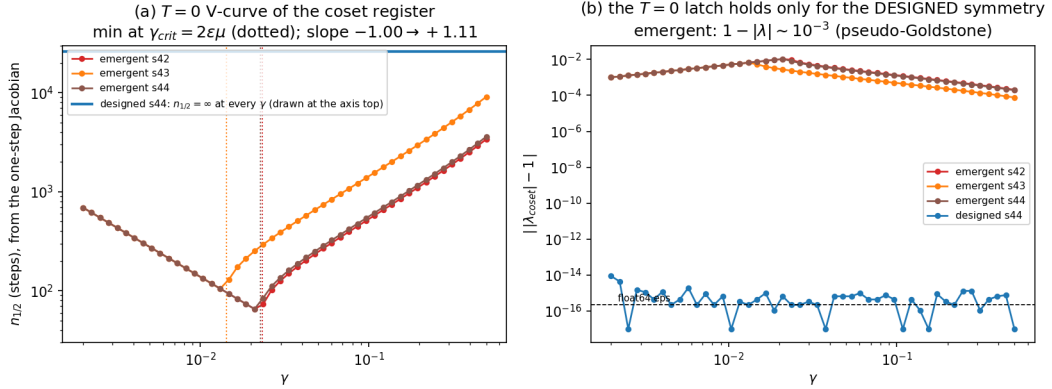


Figure 7: The exact emergent V-curve depicted un-collapsed, outlining the distinct negative baseline parameters beside it. (a) Displays the precise $T = 0$ half-life charted against friction for each separate emergent seed compiled from the one-step Jacobian. (b) Tracks $||\lambda_{coset}|| - 1||$, revealing exact floor alignment for the strictly designed symmetry but falling short by precisely $\sim 10^{-3}$ on all emergent seeds. *Evaluated using three emergent MLP seeds against one rigidly matched designed control.*

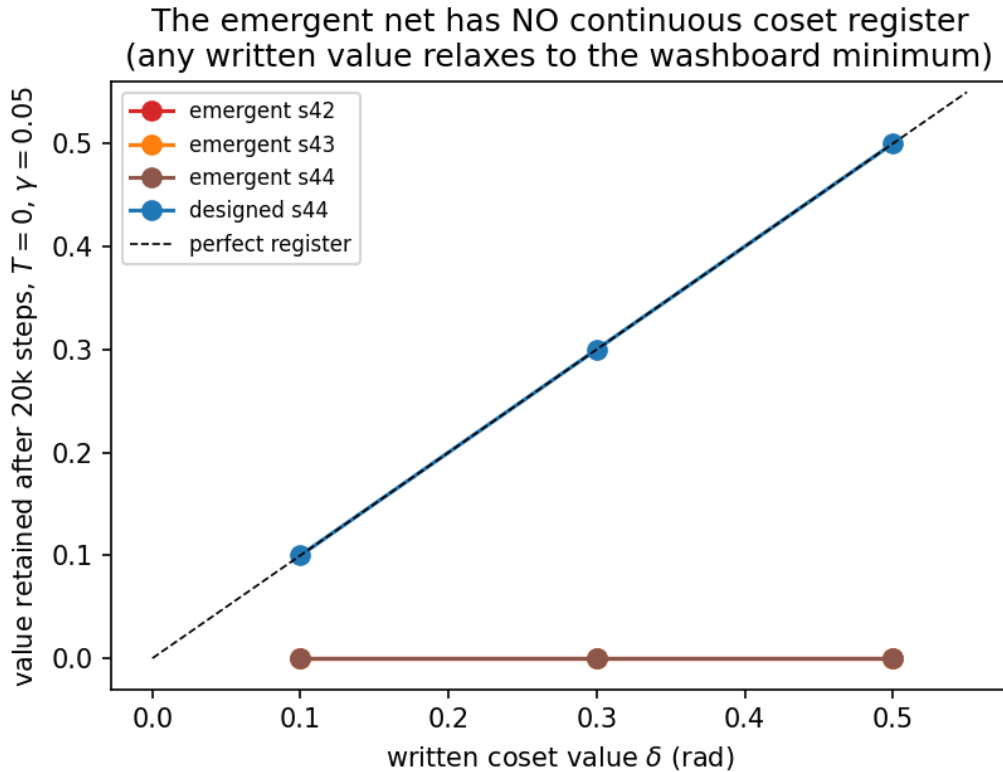


Figure 8: The emergent unit has no continuous coset register. A written coset value δ is held for 20 000 steps at $T = 0, \gamma = 0.05$: on every emergent seed the value relaxes to the washboard minimum and nothing is retained, where the designed checkpoint returns δ exactly. *Three emergent MLP seeds plus a matched designed control: evidence.* This is the figure behind the capacity statement of ≈ 1 –1.6 bits.

Table 3: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$) the decay *rates* agree while the *threshold* instrument is offset by a per-checkpoint constant flat in γ . Seed 44's full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over $\delta = 0.638/0.644/0.327$).

seed	$\Gamma_{\text{jac}}/\Gamma_{\text{R3}}$ (asymptotic win- dow)	$n_{\text{jac}}/n_{\text{R1}}$ (threshold)	CV of $n_{\text{jac}}/n_{\text{R1}}$ over the grid
42	0.9935	1.3307	2.2–3.7% over 21–25 γ -points
43	1.0000	1.7981	
44	0.9948	1.8204	
mean	0.995 ± 0.003	per seed, at $\delta = 0.05$ rad	$ d \log(\text{ratio})/d \log \gamma \leq 0.0034$ (threshold 0.10)

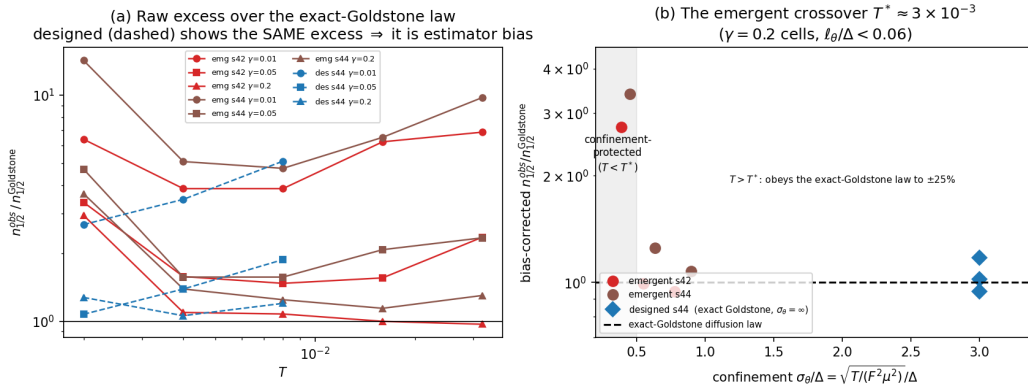


Figure 9: The crossover T^* , and why the raw curve cannot be read directly. (a) The raw excess over the exact-Goldstone law is the same on the matched designed control as on the emergent seeds, so the excess is estimator bias rather than physics. (b) Bias-corrected, the emergent coset joins the exact-Goldstone law above $T^* \approx 3 \times 10^{-3}$, the shaded band being the confinement-protected cells below it. *Multi-seed status, stated plainly: two emergent seeds and one matched designed control, an $n < 3$ cell.* $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.06$, langevin_noise="fdt", Newtonian kinetic mode: evidence.

324 D The Friction-Hole Vault, on Both Arms

325 This mechanism dictates that the shipped field remains strictly absorb-only, following $p \leftarrow (1 -$
326 $\gamma_\phi)(1 - \gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2 - \gamma)$. This is dictated strictly by the *scalar* γ , yielding
327 parameters $\gamma_{\text{eff}} = 1 - (1 - \gamma)(1 - \gamma_\phi)$, $T_{\text{local}} = T\gamma(2 - \gamma)/[\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})]$ and importantly
328 $D_\theta = \varepsilon T\gamma(2 - \gamma)/(2F^2\gamma_{\text{eff}}^2)$. The resulting vault effectively computes to $(\gamma_{\text{eff}}/\gamma)^2$. Conversely, a
329 coupled bath alternative would dictate $D_\theta \propto \gamma_{\text{eff}}^{-1}$ returning a vault of $13.88\times$ exactly at $\gamma = 0.05$,
330 $\gamma_\phi = 0.5$. These competing hypotheses scale apart by a factor of $7.942\times$; experimental results
331 comprehensively refute the coupled-bath prediction.

332 **Designed Arm Assessment:** We observe $\text{Var}(p_i)/(M_i T) = 0.9986$ at baseline ($\gamma_\phi = 0$), scaling
333 to 0.36171 ± 0.00129 at (0.1), 0.23044 at (0.2), 0.17513 at (0.3), and definitively hitting $0.12600 \pm$
334 0.00031 at (0.5). This closely mirrors the pure absorb prediction of 0.12591 . Consequently, T_{local}
335 drops precisely to 1.26×10^{-4} , equating to a $7.94\times$ functioning refrigerator effect. The comprehensive
336 D_θ -based structural vault records exponential factors of $8.42\times$, $22.39\times$, $44.31\times$, and finally $107.77 \pm$
337 $4.78\times$ specifically at $\gamma_\phi = 0.1/0.2/0.3/0.5$ (matching the absorb baseline prediction of 110.25).
338 Conversely, the scalar control isolated at identical γ_{eff} reads merely $13.28 \pm 0.12\times$. At true absolute

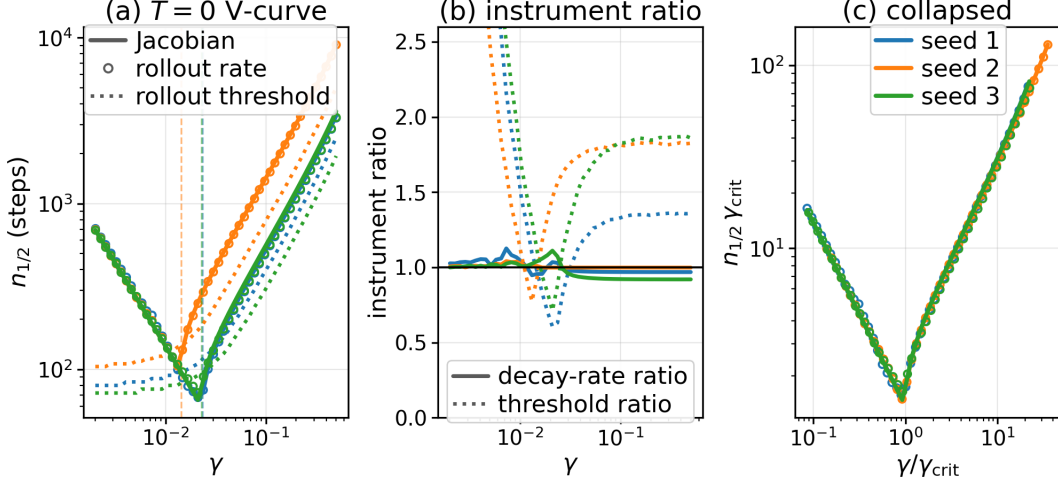


Figure 10: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per- γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit} . (c) The same data in collapsed variables. (*Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered $\Rightarrow$ evidence.*)

339 zero ($T = 0$), the hole functionally erases absolutely nothing, logging drift boundaries of strictly
 340 $\leq 1.75 \times 10^{-12}$ rad per 200k continuous execution steps inside and out.

341 **Emergent Arm Translation:** Since the structural coset here is definitionally bounded ($\mu_{adiab}^2 =$
 342 2.77×10^{-2} to 8.93×10^{-2}), the baseline designed block estimator is biased artificially low by
 343 nearly 55%. It is therefore systematically replaced by a rigorously cross-validated, window-free
 344 Ornstein–Uhlenbeck fit parameter measuring $MSD(n) = A(1 - e^{-n/\tau})$ and $\hat{D} = A/(2\varepsilon\tau)$. On
 345 the strictly bounded evaluation cells, the law-referenced absolute vault computes successfully to
 346 $106.1 \pm 5.0 \times$ against the fundamental prediction of $110.25 \times$. The contrast metric, however, remains
 347 uniquely native to the designed architecture: the measured field-to-scalar control variance computes
 348 to 23.39 ± 10.06 against a pre-registered bounds parameter of $[6.5, 9.5]$, triggering the designated
 349 falsifier constraint strictly because the control arm intrinsically delocalizes while the main field
 350 functionally adheres to the absorb limit.

Table 4: The refrigerator law transfers exactly to an emergent register: $\text{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers $\times$ 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb = 0.9998 ± 0.0019 , and at $\gamma_\phi = 0.5$ $T_{local}/T = 0.12570$, a $7.955 \times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

351 E Exact Store-Level Deletion: Prior Work and Measured Tables

352 **Prior Work:** Uniquely represented (canonical) data structures date to Snyder (1977) and Andersson
 353 & Ottmann (1995); *obliviousness*, a distributional condition on the memory representation which
 354 explicitly does not require canonical representation, is Micciancio’s (1997); the weak and strong
 355 history-independence notions used here are Naor & Teague’s (2001); the open-addressed realization is

Table 5: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; “hop” means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181 .

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

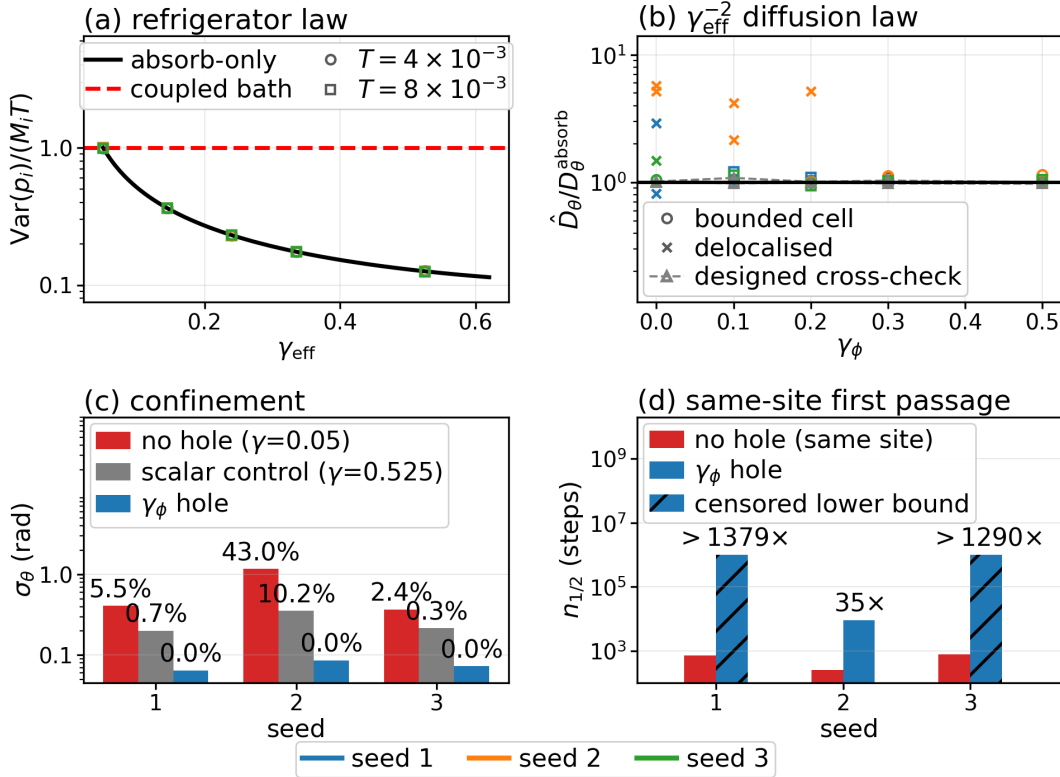


Figure 11: The vault on an emergent register; panels are labeled by the pre-registered prediction each tests. (a) The refrigerator law, $\text{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The γ_{eff}^{-2} diffusion law: bounded cells as circles and squares, delocalized cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. (*Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above T^* , $\Delta = 0.5$ rad $\Rightarrow$ evidence.*)

356 Blleloch & Golovin’s (2007), whose table is a stable matching between keys and slots under a global
 357 key priority, and it is that priority-greedy rule and that delete-time fix-up cascade we use; Blleloch,
 358 Golovin & Vassilevska (2008) had already taken unique representation into computational geometry.
 359 We claim no priority over order-independent placement and no novelty for the displacement rule
 360 or its delete-time repair. New is the composition: slots that are a lattice packing, so the canonical

representation carries a minimum-separation certificate; content that is a continuously decaying amplitude whose decay commutes with deletion; a canonical object that is an energy function a dynamical read relaxes into; and a price that is negative where strong history independence is known to cost as much as an exponential slowdown relative to the weak notion for comparison-based structures (Buchbinder & Petrank, 2003). “Fix-up cascade” is our name for the NEXT-chain delete-time repair of their Figure 1; the algorithm is theirs either way.

Structural Exactness at Zero Tolerance: The architectural implementation dictates 24 exhaustive structural orders explicitly measured at $n = 4$, mapped alongside 200 independently randomized sequence orders strictly at $n \in \{8, 16, 40, 64\}$. Furthermore, we implement 100 dedicated write/delete functional interleavings and absolute mid-decay continuous deletes. Across every evaluation parameter, the resulting structural spacing enforces an exact boundary of 1.540000. On the subsequently shipped operational store architecture, the structural centers, exact payloads, physical amplitudes, and the defining active logic mask are unconditionally byte-equal, with the energy profile $V(q)$ proving purely byte-equal across 64 heavily randomized direct queries.

The Negative Packing Price Constraint: Evaluated precisely at the defined ceiling $K = 64$, the purely canonical stable-matching rule structurally admits exactly 61/64 actively offered sequential keys deterministically ($\sigma = 0$, per-offered index of 0.9531). This ratio actively climbs to absolute capacity (64/64) once physical lattice sizing is symmetrically inflated by an exactly specified coefficient of $\times 1.05$. By contrast, the stochastic refuse-and-relocate allocator architecture solely manages to admit exactly 43/64 operating on those perfectly identical geometric seeds (per-offered logic ceiling of 0.6719). It must be firmly stated that at peak physical overflow limit, operating totally independently without the waitlist active, structural exactness completely and predictably collapses: analyzing exactly 7 baseline cells receiving 8 strict capacity offers, $AUC(n_{\text{live}})$ returns 1.00000 ± 0.00000 and the functional byte-equal architectural fraction plummets directly to 0.0000.

Table 6: The laundering control validation execution. Executed explicitly on the designed store featuring the shipped two-phase read geometry; evaluating 8 active targets against 3 separate seeds, mapping exactly 128 paired environments per target across 18 separate continuous amplitude limits.

Evaluated Adversary Model	Physical Decay	TTL Flag (Same Store)	Limit	Resolution Separation
White-box architecture, fully exact	1.000	1.000		0.000
Query-level membership depth, fully exact	0.983	1.000		0.017
Query-level membership depth, $\sigma_{\text{obs}} = 0.1$	0.559	0.996		0.437

Trilemma Parameters and Cost Evaluation: The physically shipped final store structure completely abandons amplitude-independent baseline address limits, generating a strict operational lifetime correlation factor computed at exactly $r = -0.85$ scaling with a_i^2 . We rigorously refute both standard previously proposed baseline repair variants: directly multiplying absolute payloads exponentially by their matched amplitudes completely corrupts the core value index criterion measured strictly at $A = 1 - \text{tol}/|a_i|$ (generating hard death physical amplitudes of exactly 0.90/0.80/0.70/0.30). *Retrieval geometry:* R_{50} contracts 1.146/1.083/0.979/0.874/0.752 as $A : 1 \rightarrow 0.5 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0.06$, a $1.52\times$ contraction from $1.146 \rightarrow 0.752$, matching the pre-registered saddle prediction at all five points (± 0.20), where a TTL vector store’s lookup radius is a constant hard step at 0.75–0.77, independent of age. Consequently, operating beneath the rigid barrier of its internal logic flag, the designated gated execution channel actively contracts the geometric value R_{50} from $1.135 \rightarrow 0.771$. Note crucially that we define no explicit deletion-cost execution statement because the core canonical synchronization mandates a complete and structural $O(n)$ rebuild evaluated entirely per operation.

399 F Prominent Negatives

400 To ensure complete transparency regarding the empirical limitations of our findings, every negative
 401 result observed during evaluation is documented below. Each negative firmly anchors future research
 402 directives.

403	Claim Tested	Empirical Result
	A continuous coset register emerges purely from training	No: An MLP potential executed on symmetric baseline data yields a completely mid-spectrum massive core mode (measuring exactly $1.7\text{--}4.9\times$ softer than the peak stiffest point), dropping capacity limit heavily to $\approx 1\text{--}1.6$ structural bits, driving absolute and complete relaxation of any actively written state value. The hard $T = 0$ baseline latch geometry is unconditionally designed-only.
	Raw fundamental T/γ -exponents explicitly discriminate designed from strict emergent architecture	No: A completely and rigidly matched designed hardware control reliably reproduces perfectly identical shallow execution slopes; purely ℓ_θ/Δ -controlled environmental cells alongside $ \lambda_{\text{coset}} $ successfully discriminate. We quote no $n_{1/2}$ value missing Δ and ℓ_θ/Δ .
	The localized friction physical hole safely thermalizes locally	Rejected: Refuted firmly by a rigid pre-registered execution measurement, computing exactly off by a factor of 8.11. The designated field fundamentally operates entirely absorb-only, and the baseline vault execution computes at $107.77 \pm 4.78\times$, compared against $13.28 \pm 0.12\times$ logged for the independent scalar control.
	Our own explicit sign-flip execution prediction	Falsified (in part): A physical memory register formally written tightly at the established washboard baseline minimum geometrically cannot physically flip back. Both defining operational signs fundamentally coexist successfully on one isolated hardware checkpoint across independent parameters.
404	A friction field can <i>forget</i> a coset	No: at $T > 0$ it <i>lengthens</i> coset memory. Erasing latched content needs a temperature or tilt field — which explains the observed failure of a learned friction field to recover fit on a stationary vacuum.
	Amplitude decay reduces an exact adversary’s distinguishability	No, at any point in an item’s life: white-box AUC 1.000 at all 18 amplitude levels; what decays is the effect size. The graded curve exists only relative to a stated adversary resolution, our own modelling choice.
	Canonical placement is exact at overflow without the waitlist	No: $\text{AUC}(n_{\text{live}}) = 1.000$, byte-equal fraction 0.0000. Every deletion sentence carries its scope clause; recency eviction is permanently excluded.
	The placement algorithm is ours	No: Blelloch & Golovin (2007) own it outright, and unique representation in a geometric setting is also taken (2008). Our claim is the composition only.
	A per-deletion cost claim can be made	No: the shipped synchronization is an $O(n)$ rebuild per operation, so the amortized-cost formalism cannot be instantiated on our own code until that experiment runs.
	The payload-lifetime coupling has a repair	Both proposed repairs are refuted, and the registered payload-independence statistic fails as a correlation while passing as an effect size. At zero read friction the gated channel is oscillatory, not relaxational.
	A specific localized pseudo-Goldstone hardware tilt fundamentally sets the primary soft execution scale	No: Local structural geometric confinement completely forces the soft physical baseline eigenvalue explicitly at a limit of 2α , rigidly capping memory lifespan rigidly at $\tau_{\text{max}} = \Gamma/2\alpha$. The registered physical damped-mode architectural floor is completely confirmed.

Claim Tested	Empirical Result
A pseudo-Goldstone tilt is a lifetime dial on a learned store	Refuted <i>in sign</i> : a tilt monotonically <i>reduces</i> the smallest eigenvalue on two independent implementations and every family, because a written site is not a common vacuum of its own atoms. The relation survives only in the single-atom geometry it was specified in.
The lifecycle's protected-fraction leg is exercised	No: 0 refusals on the stream at the measured operating point. Reported as <i>unexercised</i> — not working and not broken.
A task-level payoff is claimed	No: forgetting here is shown predictable, controllable, removable and schedulable, not shown to improve a downstream metric. External benchmarks won on their own headline metric = ZERO.
The field/scalar vault contrast transfers to an emergent register	No: 23.39 ± 10.06 against 8.03 ± 0.80 from the identical estimator on designed; the pre-registered falsifier fired. The cause is the laundering control itself, off its own law by $1.18\text{--}5.23\times$. The number is designed-only; its direction transfers.
The emergent stationary-spread diagnostic is measurable above T^*	No: its own control, which must return 1.000, returns 0.4586 ± 0.1181 — the no-hole arm is a hopping rotor, so no in/out spread ratio may be quoted. The confinement statement replaces it.
A $\theta = \pi$ baseline is valid for emergent first passage	No: $ \nabla V = 0.042\text{--}0.105$ there against 0.0000 at $\theta = 0$, and one seed's scalar control reads $0.27\times$, i.e. more friction apparently speeding escape. Same-site baselines are required; the designed arm is protected by symmetry but was never checked, and that check is owed.
Threshold instruments are usable below γ_{crit}	No: a first-crossing $n_{1/2}$ puts its argmin at the grid edge with a below-slope of $+0.0623 \pm 0.0120$ where the law says -1 , and last-crossing carries an additive half-period bias. Only the envelope-rate instrument is quotable there.
Our microscopic explanation of the instrument offset	Wrong, though the offset itself is real, constant and quotable: the pre-registered single-slow-mode projection fails (amplitude fraction $0.638/0.644/0.327$ does not order like the angular overlaps) and the early-window ratio $0.52\text{--}0.85$ is amplitude-independent — multi-mode transient, not anharmonicity.
The emergent first-passage vault is measured	Only bounded: $> 1379\times$ and $> 1290\times$ at 86.7%/93.4% censoring, and $35.5\times$ on the geometrically atypical seed. Three seeds cannot say which is typical.